# MNIST CNN ReLU Clipping Sweep Report: No Clip vs. Clip 1/2/4 vs. Tanh vs. Grad-Norm Clip 21

ByzFL SNN Benchmark

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# Contents

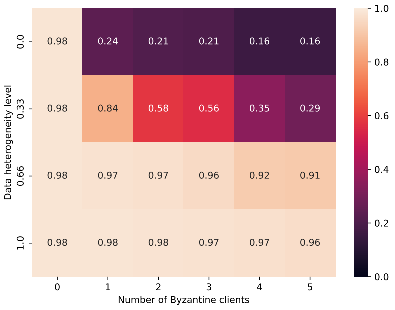
# 1 Introduction

This report compares six MNIST CNN configurations:

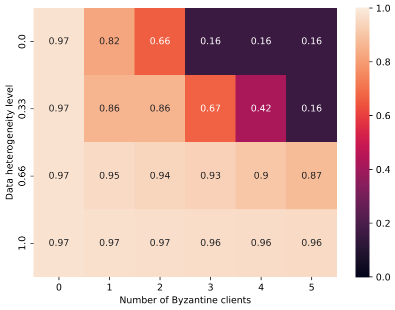
1. **CNN ReLU (No Clip)**: Standard CNN with ReLU, no activation clipping.
2. **CNN ReLU (Clip** $= 1$**)**: ReLU activations clipped elementwise to $[0, 1]$.
3. **CNN ReLU (Clip** $= 2$**)**: ReLU activations clipped elementwise to $[0, 2]$.
4. **CNN ReLU (Clip** $= 4$**)**: ReLU activations clipped elementwise to $[0, 4]$.
5. **CNN Tanh**: CNN with Tanh activation (implicit saturation).
6. **CNN ReLU (Grad-Norm Clip** $= 21$**)**: ReLU network with the gradient's overall norm clipped to 21 (not a per-neuron/elementwise clip).

**Note:** The Grad-Norm Clip = 21 configuration only has heatmaps available for the best-aggregator (worst-case) sections and the Geometric Median sections.

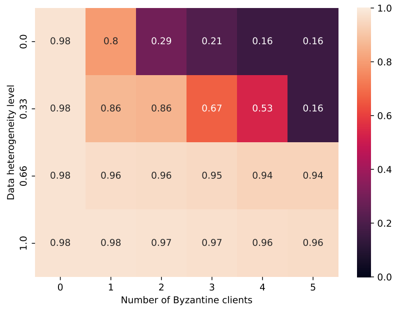
# 2 Best Overall Performance (Worst-Case Across Attacks)



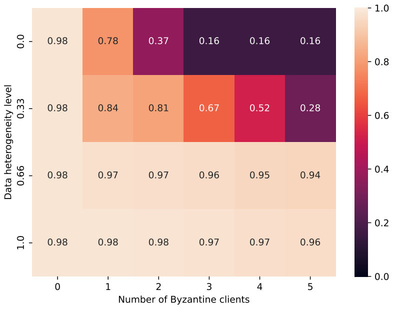

(a) CNN ReLU (No Clip)



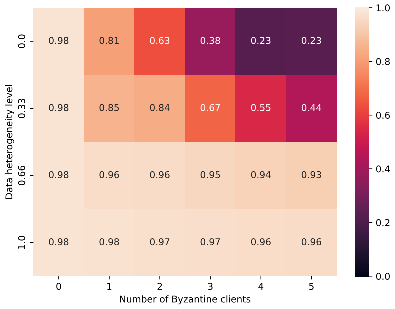

(b) CNN ReLU (Clip = 1)



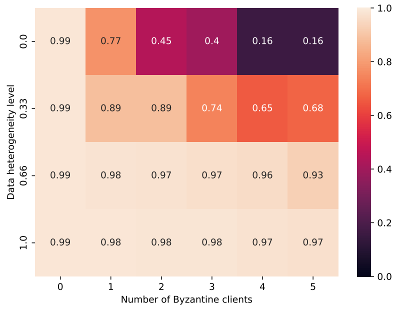

(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)




(e) CNN Tanh

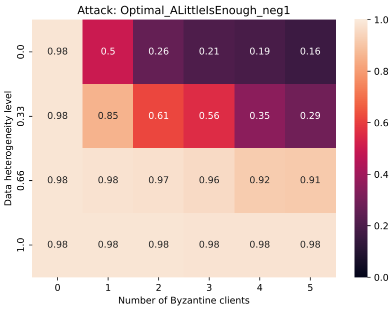



(f) CNN ReLU (Grad-Norm Clip = 21)

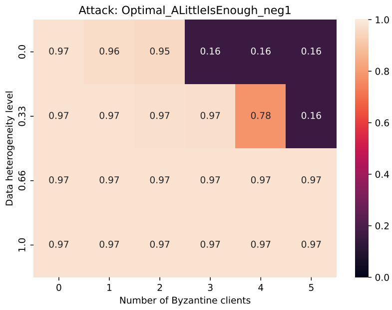
Figure 1: Best overall test accuracy (worst-case across attacks).

# 3 Best Aggregator under Specific Attacks

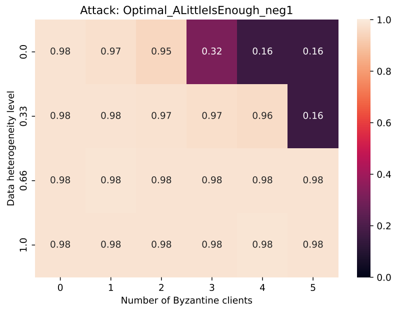
## 3.1 Optimal ALIE Attack



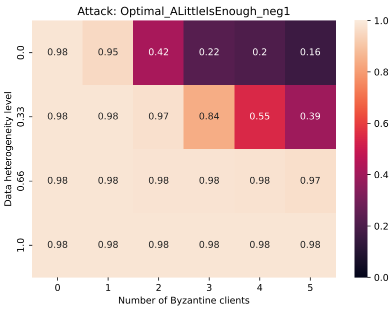

(a) CNN ReLU (No Clip)



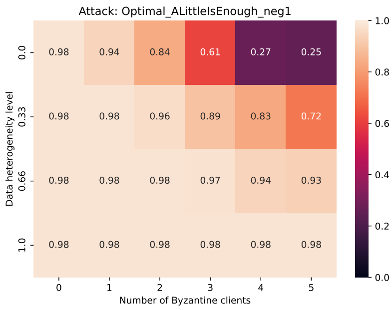

(b) CNN ReLU (Clip = 1)



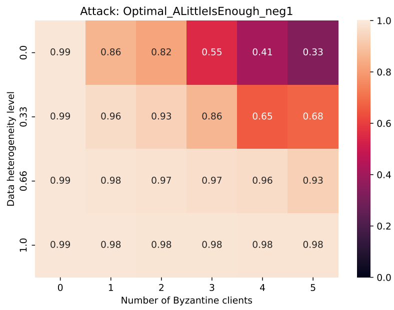

(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)



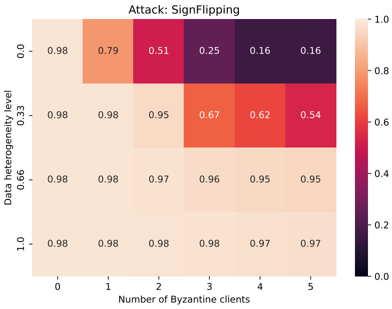

(e) CNN Tanh



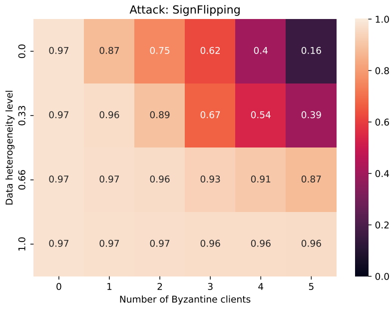

(f) CNN ReLU (Grad-Norm Clip = 21)

Figure 2: Best test accuracy under ALIE.

## 3.2 Sign Flipping Attack




(a) CNN ReLU (No Clip)




(b) CNN ReLU (Clip = 1)


(c) CNN ReLU (Clip = 2)

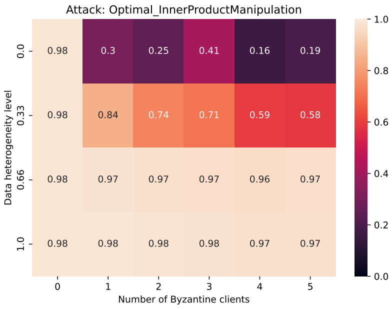

(d) CNN ReLU (Clip = 4)


(e) CNN Tanh

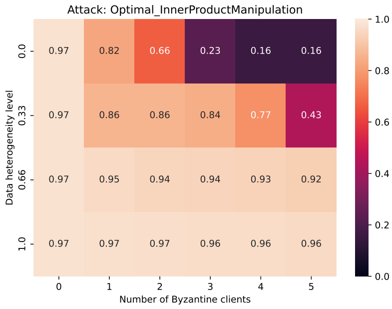

(f) CNN ReLU (Grad-Norm Clip = 21)

Figure 3: Best test accuracy under SF.

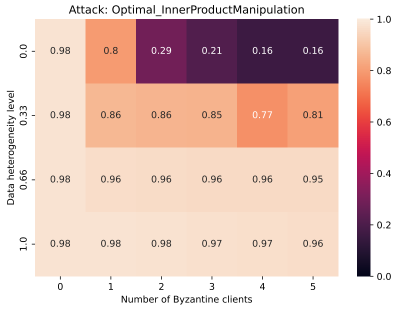
## 3.3 Optimal IPM Attack



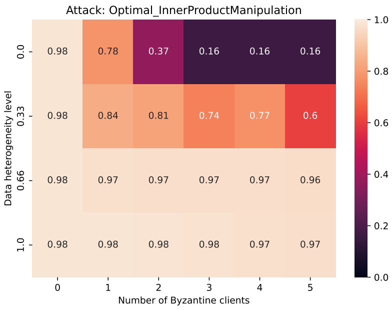

(a) CNN ReLU (No Clip)



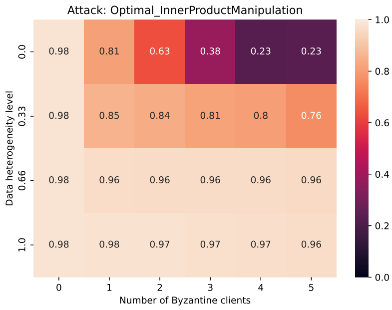

(b) CNN ReLU (Clip = 1)



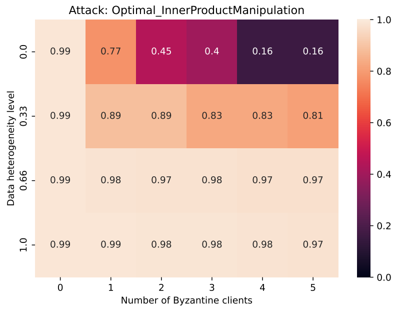

(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)




(e) CNN Tanh

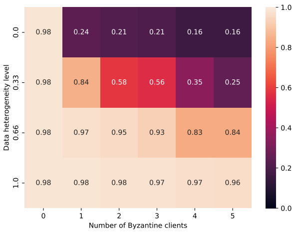



(f) CNN ReLU (Grad-Norm Clip = 21)

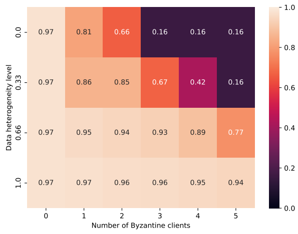
Figure 4: Best test accuracy under IPM.

## 3.4 Centered Clipping (CC)

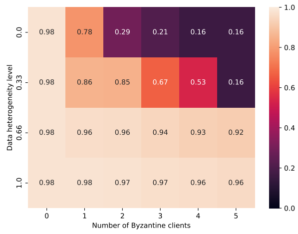
### 3.4.1 CC under Worst-Case across all Attacks



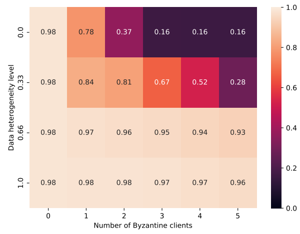

(a) CNN ReLU (No Clip)




(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)

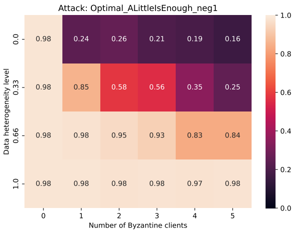



(d) CNN ReLU (Clip = 4)

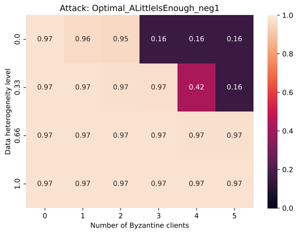

(e) CNN Tanh

Figure 5: Centered Clipping (CC) under Worst-Case across all Attacks.

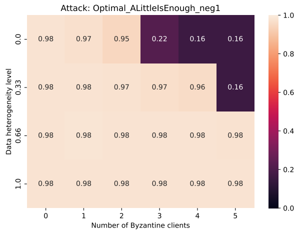
### 3.4.2 CC under ALIE



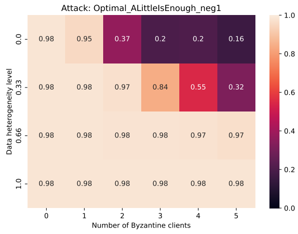

(a) CNN ReLU (No Clip)



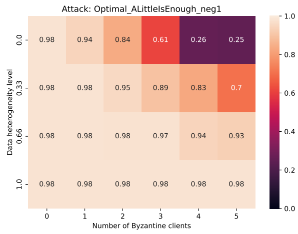

(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)

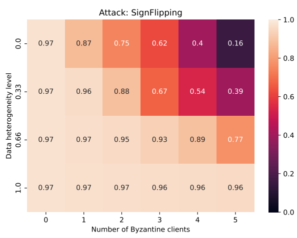



(e) CNN Tanh

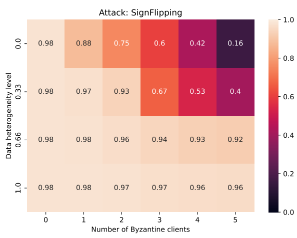
Figure 6: Centered Clipping (CC) under Optimal ALIE Attack.

### 3.4.3 CC under SF

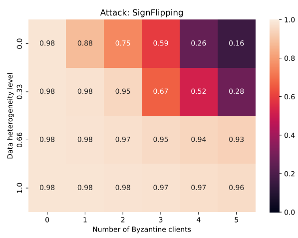

(a) CNN ReLU (No Clip)



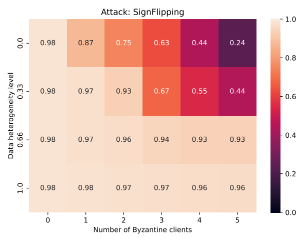

(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)



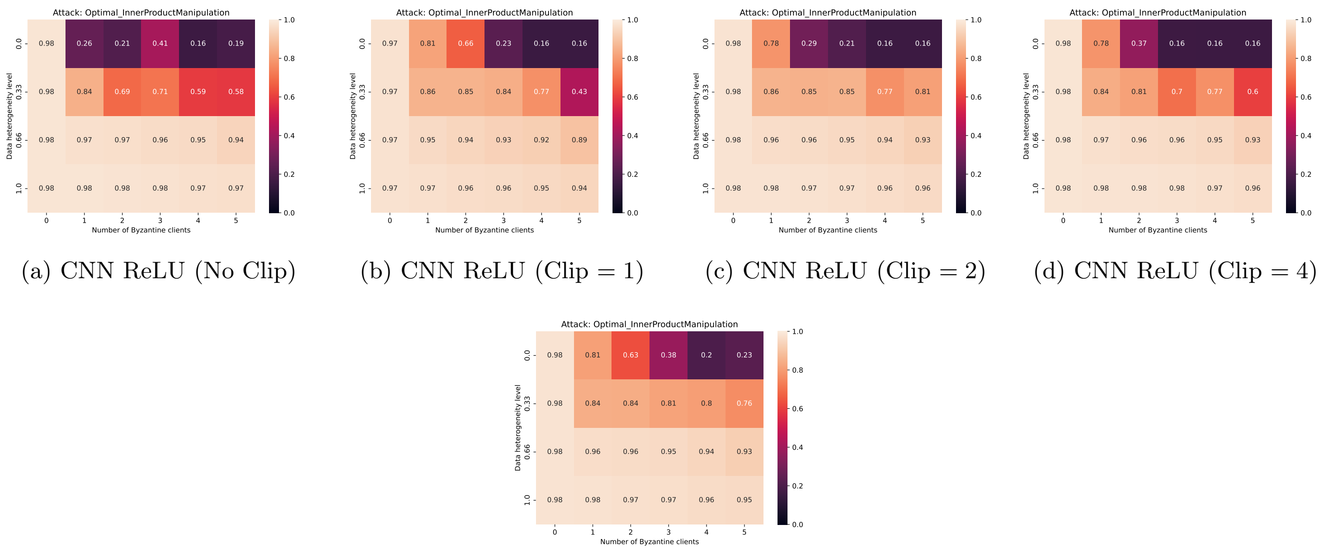

(d) CNN ReLU (Clip = 4)




(e) CNN Tanh

Figure 7: Centered Clipping (CC) under Sign Flipping Attack.

### 3.4.4 CC under IPM




(a) CNN ReLU (No Clip)

(b) CNN ReLU (Clip = 1)

(c) CNN ReLU (Clip = 2)

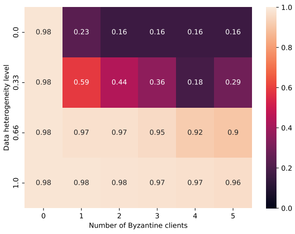
(d) CNN ReLU (Clip = 4)

(e) CNN Tanh

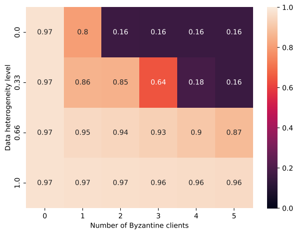
Figure 8: Centered Clipping (CC) under Optimal IPM Attack.

## 3.5 Geometric Median (GM)

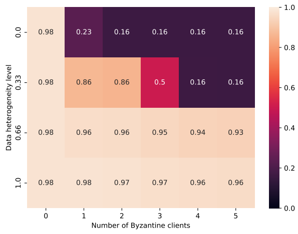
### 3.5.1 GM under Worst-Case across all Attacks




(a) CNN ReLU (No Clip)



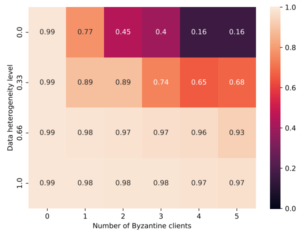

(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)


(d) CNN ReLU (Clip = 4)

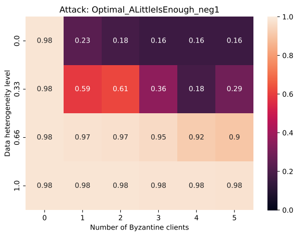

(e) CNN Tanh



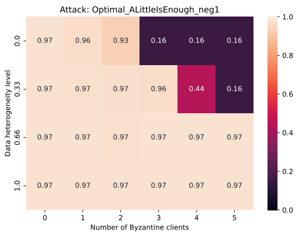

(f) CNN ReLU (Grad-Norm Clip = 21)

Figure 9: Geometric Median (GM) under Worst-Case across all Attacks.

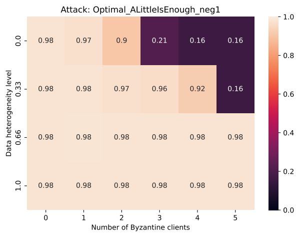
### 3.5.2 GM under ALIE



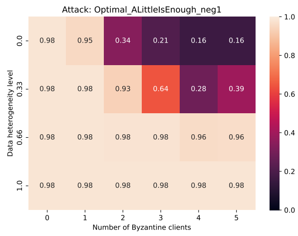

(a) CNN ReLU (No Clip)



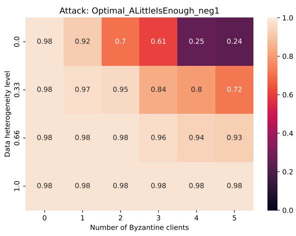

(b) CNN ReLU (Clip = 1)



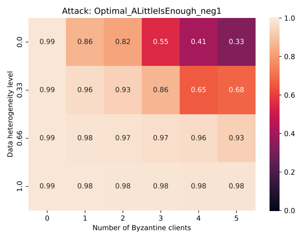

(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)



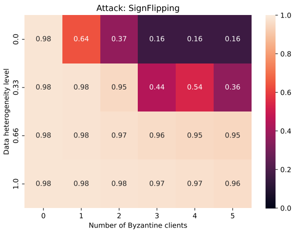

(e) CNN Tanh




(f) CNN ReLU (Grad-Norm Clip = 21)

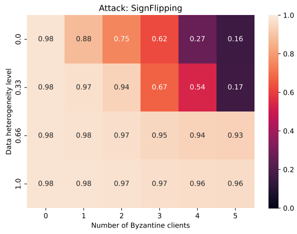
Figure 10: Geometric Median (GM) under Optimal ALIE Attack.

### 3.5.3 GM under SF

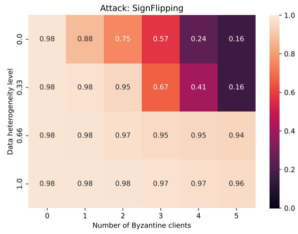



(a) CNN ReLU (No Clip)

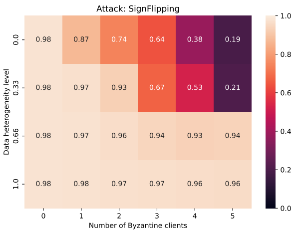

(b) CNN ReLU (Clip = 1)



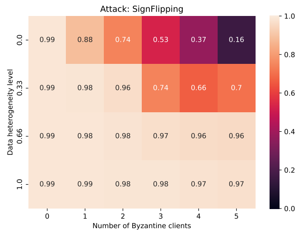

(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)



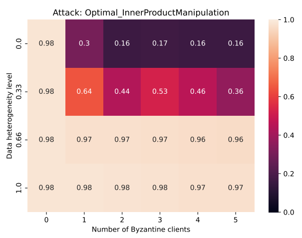

(e) CNN Tanh



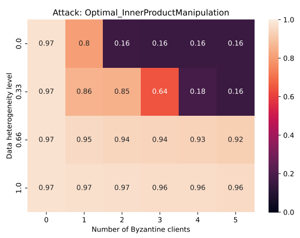

(f) CNN ReLU (Grad-Norm Clip = 21)

Figure 11: Geometric Median (GM) under Sign Flipping Attack.

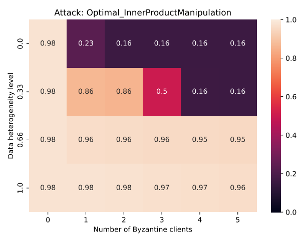
### 3.5.4 GM under IPM



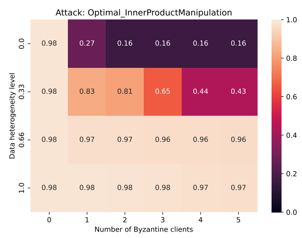

(a) CNN ReLU (No Clip)



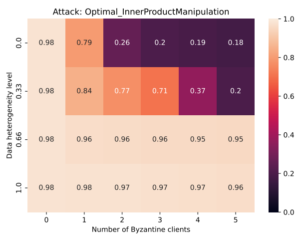

(b) CNN ReLU (Clip = 1)



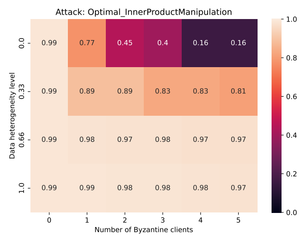

(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)




(e) CNN Tanh

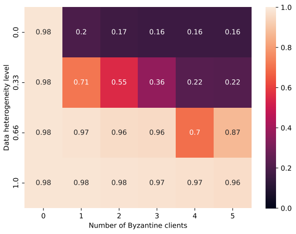



(f) CNN ReLU (Grad-Norm Clip = 21)

Figure 12: Geometric Median (GM) under Optimal IPM Attack.

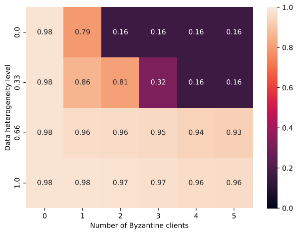
## 3.6 Multi-Krum (MK)

### 3.6.1 MK under Worst-Case across all Attacks




(a) CNN ReLU (No Clip)


(b) CNN ReLU (Clip = 1)



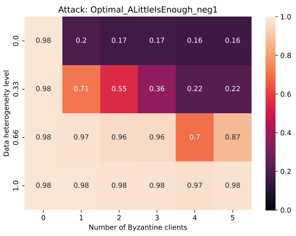

(c) CNN ReLU (Clip = 2)


(d) CNN ReLU (Clip = 4)

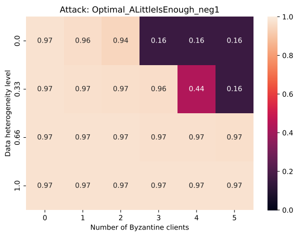

(e) CNN Tanh

Figure 13: Multi-Krum (MK) under Worst-Case across all Attacks.

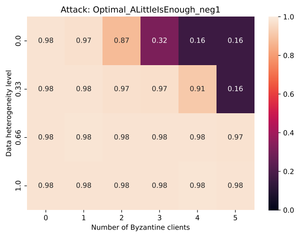
### 3.6.2 MK under ALIE



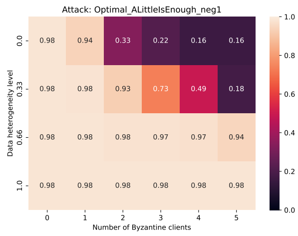

(a) CNN ReLU (No Clip)



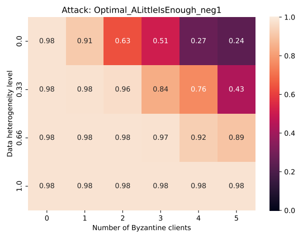

(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)



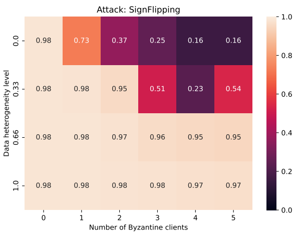

(d) CNN ReLU (Clip = 4)



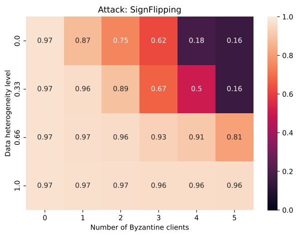

(e) CNN Tanh

Figure 14: Multi-Krum (MK) under Optimal ALIE Attack.

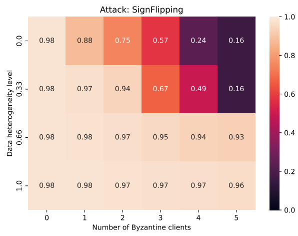
### 3.6.3 MK under SF



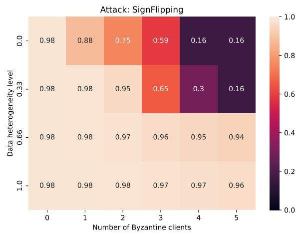

(a) CNN ReLU (No Clip)




(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)

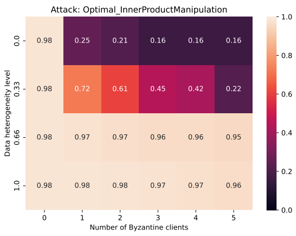



(d) CNN ReLU (Clip = 4)

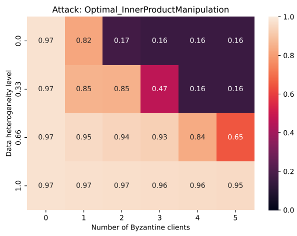

(e) CNN Tanh

Figure 15: Multi-Krum (MK) under Sign Flipping Attack.

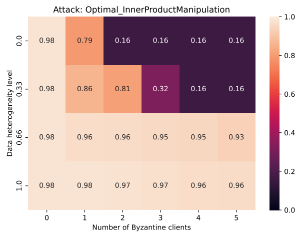
### 3.6.4 MK under IPM



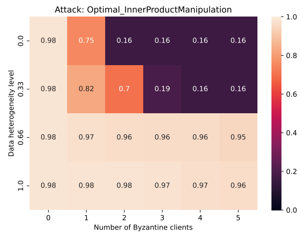

(a) CNN ReLU (No Clip)



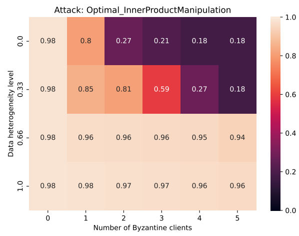

(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)

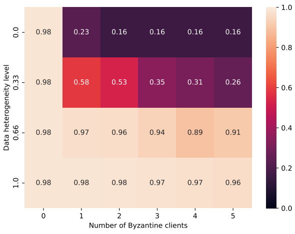



(e) CNN Tanh

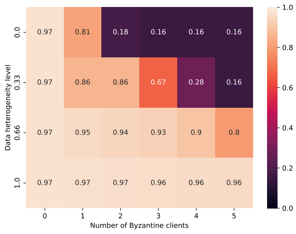
Figure 16: Multi-Krum (MK) under Optimal IPM Attack.

## 3.7 Trimmed Mean (TM)

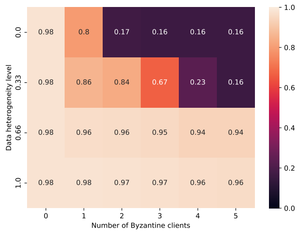
### 3.7.1 TM under Worst-Case across all Attacks




(a) CNN ReLU (No Clip)

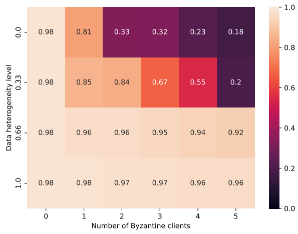



(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)

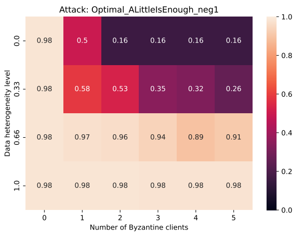

(d) CNN ReLU (Clip = 4)



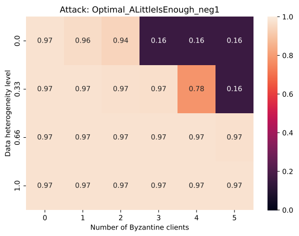

(e) CNN Tanh

Figure 17: Trimmed Mean (TM) under Worst-Case across all Attacks.

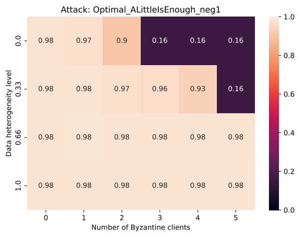
### 3.7.2 TM under ALIE



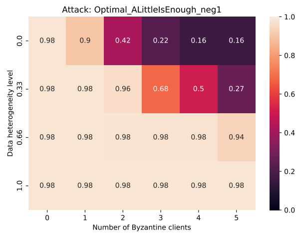
(a) CNN ReLU (No Clip)



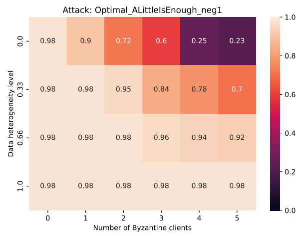
(b) CNN ReLU (Clip = 1)



(c) CNN ReLU (Clip = 2)



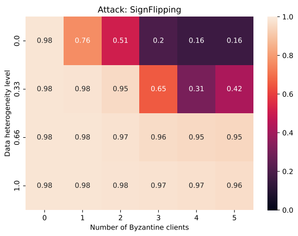
(d) CNN ReLU (Clip = 4)



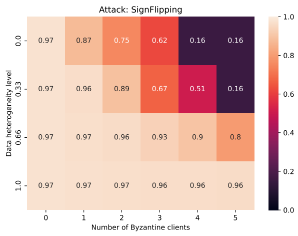
(e) CNN Tanh

Figure 18: Trimmed Mean (TM) under Optimal ALIE Attack.

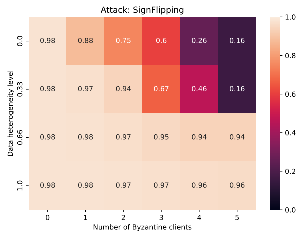
### 3.7.3 TM under SF




(a) CNN ReLU (No Clip)

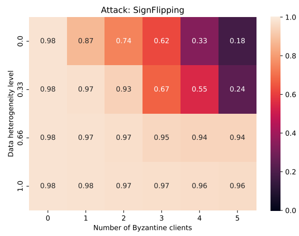



(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)

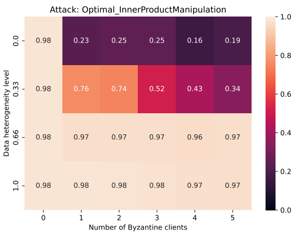

(d) CNN ReLU (Clip = 4)



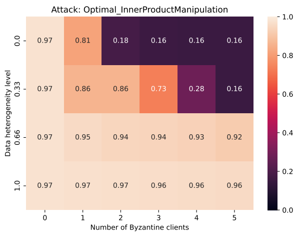

(e) CNN Tanh

Figure 19: Trimmed Mean (TM) under Sign Flipping Attack.

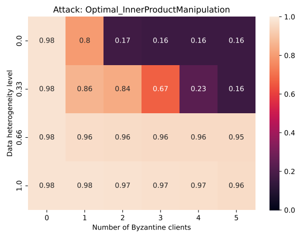
### 3.7.4 TM under IPM



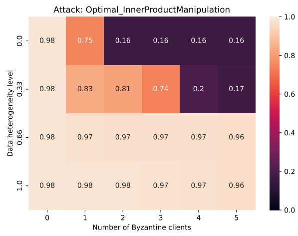

(a) CNN ReLU (No Clip)



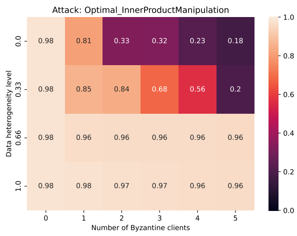

(b) CNN ReLU (Clip = 1)




(c) CNN ReLU (Clip = 2)




(d) CNN ReLU (Clip = 4)




(e) CNN Tanh

Figure 20: Trimmed Mean (TM) under Optimal IPM Attack.